

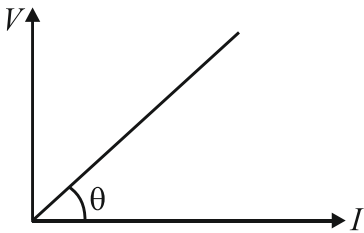
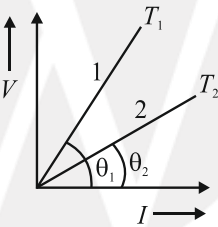


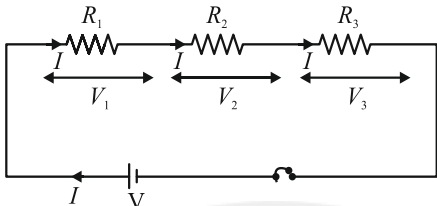
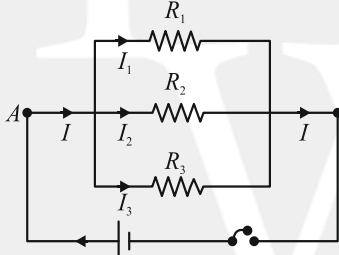
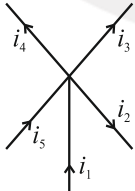
CHAPTER

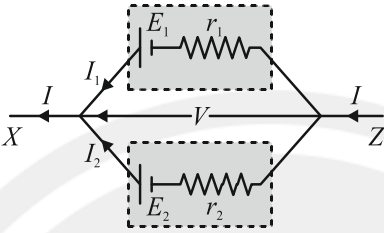
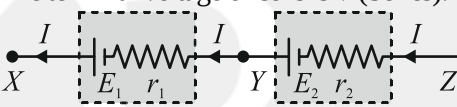
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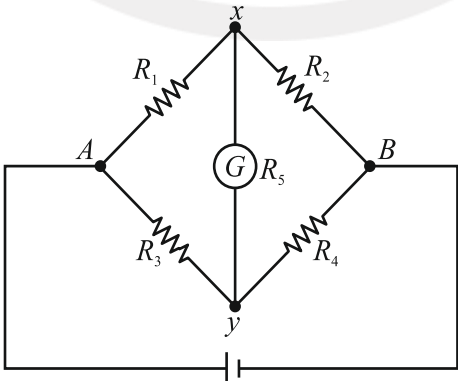
Current Electricity

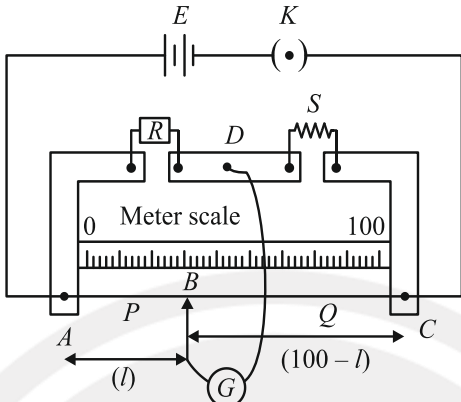
Sr. No.	Concept	Formulas	Other information
1.	Electric Current	$I = \frac{\text{Total charge flowing } (q)}{\text{Total time taken } (t)}$ $\Rightarrow I = \frac{q}{t} = \frac{ne}{t}$ Total charge flow in time interval t_1 to t_2 can be given as $Q = \int_{t_1}^{t_2} I \cdot dt$	n = number of carriers (electrons) e = charge of electron
2.	Average and instantaneous current	Average current is defined as $I_{av} = \frac{\Delta q}{\Delta t}$ Instantaneous current $I_{inst} = \lim_{\Delta t \rightarrow 0} \frac{\Delta q}{\Delta t} = \frac{dq}{dt}$	dq = charge flows through a conductor in small time dt .
3.	Current in the case when charge q is revolving in a circle	$i = \frac{q}{t} = qf = \frac{q}{\frac{2\pi}{\omega}} = \frac{qv}{2\pi r}$	q = charge revolving in a circle of radius r with uniform velocity v . ω = the angular frequency
4.	Current density (J)	$J = \frac{I}{A}$ If area is finite, $I = \int \vec{J} \cdot d\vec{A}$	dA = small element of the given surface I = the current flowing through the surface area A
5.	Drift velocity (V_d) of electrons in conductors	$v_d = \frac{eE}{m} \tau, i = neAv_d, v_d = \frac{i}{neA} = \frac{J}{ne}$	e = charge of electron, E = electric field m = mass of the electron τ = relaxation time. A = area of cross-section, n = number density of electrons.
6.	Relaxation time (τ) and Mobility (μ)	$\tau = \frac{\text{mean free path}}{\text{rms velocity of electrons}} = \frac{\lambda}{v_{rms}}$ $\mu = \frac{v_d}{E} = \frac{e\tau}{m}$	m = mass of the electron λ = mean free path

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7.	Ohm's Law	$V \propto I$ [Ohm's law] $\frac{V}{I} = \text{constant} = R$ (Resistance) $\Rightarrow V = IR$  $\text{Slope} = \frac{V}{I} = \tan \theta = R$ The microscopic form of ohm's law $J = nev_d \Rightarrow J = \sigma E$	σ = electrical conductivity of the material. R = resistance measured V = potential difference
8.	Resistance of a conductor and resistivity	$R \propto \frac{l}{A} \Rightarrow R = \rho \frac{l}{A}$ $R = \frac{dV}{dI}$ [for non-ohmic materials] Where $\rho = \left(\frac{m_e}{ne^2\tau} \right)$, Resistivity Reciprocal of resistivity is called conductivity (σ), i.e. $\sigma = \frac{1}{\rho}$ Here $\tan \theta_1 > \tan \theta_2$ So $R_1 > R_2$ i.e., $T_1 > T_2$ 	ρ = resistivity of conductor σ = conductivity of conductor $\rho_{\text{alloy}} > \rho_{\text{semiconductor}} > \rho_{\text{conductor}}$ l = length of conductor A = area of cross section
9.	Effect of stretching on resistance of the wire.	$R \propto l^2$ If $l_2 = nl_1$ $\frac{R_2}{R_1} = \left(\frac{l_2}{l_1} \right)^2 = \left(\frac{nl_1}{l_1} \right)^2 = n^2 \quad R_2 = n^2 R_1$	l = length of the conductor. n = length of the conductor is increased by n times. On stretching = $l \uparrow$, $A \downarrow$, $R \uparrow$, Volume = Al (remains constant)
10.	Temperature dependence of resistance	Resistance at temperature $T^\circ\text{C}$ is given as $R_T = R_0(1 + \alpha\Delta T)$	R_T = resistance at $T^\circ\text{C}$, R_0 = resistance at 0°C ΔT = change in temperature, α = temperature coefficient of resistance

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11.	Series combination of resistors	In series, $I \rightarrow$ same $V = V_1 + V_2 + V_3$ $IR_{eq} = IR_1 + IR_2 + IR_3$ $R_{eq} = R_1 + R_2 + R_3$ For n identical resistances in series: $R_{eq} = nR, R_{eq} > (R_{\max})_{\text{combination}}$ 	V = P.D. of source V_1, V_2, V_3 = potential drop across each resistor R_1, R_2 and R_3 respectively R_{eq} = equivalent resistance.
12.	Parallel combination of resistors	$I = I_1 + I_2 + I_3$, For n resistors in parallel $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$ In case of 2 resistance $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R = \frac{R_1 R_2}{R_1 + R_2}$ If n equal resistance are connected in parallel $R_{eq} = \frac{R}{n}, R_{eq} < (R_{\min})$ 	V = same potential across all resistors I_1, I_2, I_3 = current flowing through the resistors R_1, R_2 and R_3. $R_{\min}$ = minimum resistance from the given resistances.
13.	Kirchhoff's laws	Junction rule $\sum i = 0$ [Kirchhoff's current law] $i_1 - i_2 - i_3 - i_4 + i_5 = 0$  Loop rule. i.e. $\sum \Delta V = 0$ [Kirchhoff's voltage law]	i_1, i_2, i_3, i_4, i_5 = current flowing through all the connected lines i = sum of all the currents.
14.	Terminal potential difference of cell (V)	Current $I = \frac{E}{r + R}$ $\Rightarrow E = IR + Ir = V + Ir$ $\Rightarrow V = E - Ir$	E = emf of cell r = internal resistance connected with a external resistance R.

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15.	Different cells in parallel and series	The terminal voltage of cells is V (Parallel): $V_{XZ} = \left(\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \right) - I \left(\frac{r_1 r_2}{r_1 + r_2} \right)$ Equivalent emf and resistance (Parallel): $E_{eq} = \left(\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \right)$ $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$  The terminal voltage of cells is V (Series):  $V_{XZ} = (E_1 + E_2) - I(r_1 + r_2)$ Equivalent emf and resistance (Series): $E_{eq} = E_1 + E_2 + \dots + E_n$ $r_{eq} = r_1 + r_2 + \dots + r_n$	E_1, E_2 = emf of the cells, r_1, r_2 = internal resistances of the cells. E_{eq} = equivalent emf after connecting all the cells. R_{eq} = equivalent resistance after connecting all the cells. I = amount of the current flowing.
16.	Grouping of cells (Series)	(i) Equivalent emf of the combination, $E_{eq} = nE$. (ii) Equivalent internal resistance of the combination, $r_{eq} = nr$. (iii) Main current = Current from each cell $= I = \frac{nE}{R + nr}$.	n = number of identical cells of emf E and internal resistance r each are connected in series. R = external resistance.
17.	Grouping of cells (Parallel)	(i) Equivalent emf of the combination, $E_{eq} = E$. (ii) Equivalent internal resistance, $R_{eq} = r/n$. (iii) Main current, $I = \frac{E}{R + r/n}$.	V = potential difference. r = internal resistance. E = emf of the equivalent cells.
18.	Mixed grouping of cells	(i) Equivalent emf of the combination, $E_{eq} = nE$ (ii) Equivalent internal resistance, $r_{eq} = \frac{nr}{m}$ (iii) Main current flowing through the load, $I = \frac{nE}{R + \frac{nr}{m}} = \frac{mnE}{mR + nr}$	n = number of identical cells of emf E and internal resistance r are connected in a row and such m rows are connected in parallel.

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19.	Electrical energy (Joule's law) and power	Electrical Power: $P = \frac{W}{t} = VI = I^2 R = \frac{V^2}{R}$ Heat produced: $H = I^2 R t = \frac{V^2 t}{R} = VIt$	1 HP = 746 Watt. H = amount of heat generated. W = total workdone on electric charge. t = time.
20.	Power consumption in a combination of bulbs	In Series: $P_{\text{consumed}} (\text{brightness}) \propto V \propto R \propto \frac{1}{P_{\text{rated}}}$ In parallel: $P_{\text{consumed}} (\text{brightness}) \propto P_{\text{rated}} \propto I \propto \frac{1}{R}$ When two bulbs are connected in series $P = \frac{P_1 P_2}{P_1 + P_2}$ When two bulbs are connected in parallel $P = P_1 + P_2$	P_1, P_2 = power of the two bulbs P = total power consumed. V = potential applied. R = resistance of the bulbs.
21.	Maximum power transfer theorem	$\Rightarrow P = \left(\frac{E}{R+r} \right)^2 \times R$ $P \text{ will be maximum if } \frac{dP}{dR} = 0$ $\Rightarrow r^2 - R^2 = 0 \Rightarrow r = R, P_{\text{max}} = \frac{E^2}{4R}$	E = emf of the cell. P_{max} = maximum power delivered r = internal resistance.
22.	Wheatstone bridge	Balancing condition $\frac{R_1}{R_2} = \frac{R_3}{R_4}$ $R_1 R_4 = R_2 R_3$ If $\frac{R_1}{R_2} < \frac{R_3}{R_4}$; then $V_x > V_y$ If $\frac{R_1}{R_2} > \frac{R_3}{R_4}$; $V_x < V_y$ 	V_x, V_y = potentials of the points x and y respectively. G = galvanometer.

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23.	Meter bridge	At balancing situation of bridge, $\frac{P}{Q} = \frac{R}{S}$ $\frac{l}{100-l} = \frac{R}{S}, S = \frac{100-l}{l} \times R$ 	S = unknown resistance. l = balancing length of the bridge wire. R = known resistance.
24.	Potentiometer and its uses	The potentiometer equation is $\frac{E_1}{E_2} = \frac{l_1}{l_2}$. Potential Gradient $\lambda = \frac{V}{L} = I \left(\frac{R}{L} \right)$ Internal resistance of the cell: $r = \left(\frac{l_1 - l_2}{l_2} \right) \cdot R$	E_1, E_2 = emfs of two cells is to be compared. l_1, l_2 = balancing lengths without shunt and with the shunt respectively. R = shunt resistance in parallel with the given cell.